

Study of Different Silicon Physical Unclonable Functions

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Abstract– Physical Unclonable Function (PUF) provides, efficient and cheap authentication. PUF is a device that harnesses the entropy of a physical system. A PUF will generate a specific bit pattern response for a specific challenge. But different instances of the same PUF circuit will produce different bit patterns. Since the response depends on the inherent properties, the attacker cannot replicate the PUF device even by knowing its design. A detailed survey of different types of Silicon PUF designs has been done and a new delay based Silicon PUF design has been proposed in this paper.

KEYWORDS- Physical Unclonable Function (PUF), Process variations, Delay elements, Reliability, Uniqueness, Uniformity.

I. INTRODUCTION

In today's world, IC security is an important area of concern. We can identify a number of hardware design threats such as

- Counterfeiting – Copying the same IC. It is also known as cloning by which attacker can copy and resell a design.
- Reverse engineering – Extracting IP or sensitive data, such as cryptographic keys, in the on-chip memory. It is a threat to intellectual property and data security.
- Tampering to IC operation or insert malicious functionality, using Trojans or kill switches. Tampering is the attempts of an attacker to gain unauthorized access to an electronic system. It is a threat to integrity and trustworthiness.

In order to overcome such threats, the physical unclonable functions or PUFs are an alternate option.

Using the built-in physical properties of a system for identification and security is not a new idea. The identification of people using fingerprints started in the 19th century. In the same way, intrinsic properties of a circuit can be used to identify and secure the integrated circuits. PUF are also known as physically unclonable functions. PUFs are the physical entity which is substantiated on a physical structure. The PUF's response is nearly impossible to predict. An individual PUF device should be easily manufacturable, but practically it should be impossible to clone even with the same manufacturing process [13].

A. Challenge-Response Pair (CRP)

A PUF generally executes a functional operation by a circuit. A challenge is an input given to PUF and response is the output for the corresponding input. A specific challenge and its matching response together termed as Challenge-Response Pair (CRP).

B. PUF Properties

Main PUF properties include reliability, uniformity and uniqueness.

- Reliability: For particular challenges with reference to varying temperature and supply voltage, the response of the PUF should be consistent. The extent to which a PUF deviates from its reliability is known as the error rate. The error rate can be calculated from the average intra-chip hamming Distance (HD) between a group of samples, for a specific PUF instance i . Based on a group of intra-chip HD, expression to get a single value the calculations is as shown below [2].

$$HD_{INTRA} = \frac{1}{m} \sum_{t=1}^m \frac{HD(R_i, R'_{i,t})}{n} \times 100\%$$

The average numbers of unreliable or noisy PUF response bits are indicated by HD_{INTRA} . Reliability can be expressed as

$$\text{Reliability} = 100\% - HD_{INTRA}$$

- Uniformity: Uniformity implies 1's and 0's in the response should be equally distributed. It is alternatively known as randomness. An ideal PUF shows 50% of uniformity. It is effectively the mean value of an n -bit response, and can be expressed as follows. [2]

$$U_i = \frac{1}{n} \sum_{l=1}^n r_{i,l} \times 100\%$$

- Uniqueness: The responses to a particular challenge should vary, for two instances of the same PUF design. The creation of unique keys or signatures demands this property. A PUF duplicated on a different chip should generate a signature with HD of around fifty percentages, which implies half the bits are totally different. For a group of k chips, the uniqueness of a PUF can be calculated using given equation [2].

$$\text{Uniqueness} = \frac{2}{k(k-1)} \sum_{i=1}^{k-1} \sum_{j=i+1}^k \frac{HD(R_i, R_j)}{n} \times 100\%$$

C. Advantages of PUF

- Generating volatile and unique signatures for ICs at runtime is a necessary advantage of Physical Unclonable Function (PUF).
- Key storage does not require expensive and non-volatile memory.
- PUFs provide a strong security shield against attacks.

D. Applications of PUF

PUFs have many applications including

- PUFs for system Identification

Identification Numbers can be used to uniquely identify the ICs. Similar to a biometrical identification scheme, by using PUF responses identification can be done directly. From an individual PUF of a population, a large range of CRPs is kept in the database along with ID of the physical system implanting the PUF, in the enrollment phase. The verifier selects a CRP of his choice from the CRPs kept in the database of the system identification in the identification phase. The identification is successful if the obtained response and the output within the database for a specific input are identical.

- PUFs for secret key generation

The key is produced from intrinsic and inherent entropy caused by fabricating process variations; no specific key-programming step is needed. Since this randomness is built-in for a chip, no standard non-volatile storage is needed. This assures extra protection against the side channel and probing attacks. The PUF is often used to offer tamper-proof key storage.

- PUFs in Random Number Generation

A PUF design with some modification can also be used as a real, or cryptographically secure, random number generator. True random number generators are created by exploiting D-Flip flop metastability, Ring Oscillators, and SRAM PUFs.

- PUFs for Cryptography

Another application of PUF is in the generation of secrets for cryptography. The benefit of a PUF is, these secrets are not to be kept on the hardware because they are generated dynamically at device reset. PUFs will prevent the vulnerabilities such as copying or reverse engineering since physically disassembling of such a circuit can destroy its delay characteristics and thus modify its output.

E. Design Goals of PUF

Design goals of PUF include

- PUF area Minimization.
- Dynamic power consumption Minimization.
- Evaluation time Minimization.

II. LITERATURE SURVEY

A lot of new PUF concepts appeared in the past fifteen years. The PUF classification can be done based on

manufacturing material or depends on the number of CRPs. [14]

PUFs can be divided into strong and weak PUFs, based on the number of CRPs. The strong PUF includes the optical PUF, arbiter PUF, lightweight secure PUF, etc. The weak PUF mainly includes the memory-based PUF, RO PUF and glitch PUF.

Based on the material of manufacturing, the PUFs can be divided into Silicon and non-Silicon PUFs. The Non-Silicon PUFs mainly includes the optical PUF, acoustical PUF, paper PUF, RF-DNA PUF, CD PUF, magnetic PUF, and phosphor PUF. In this paper, a literary review has been performed to obtain a good idea about different types of Silicon PUFs.

Silicon PUFs

Silicon PUFs are more interesting in terms of manufacturing cost and readiness in integrating with computing and communication devices. Silicon PUFs utilize the uncontrollable manufacturing process variations to produce a novel signature for every chip. According to the different sources of variation, there are three major silicon PUFs available. They are memory-based PUFs, analog electronic PUFs, and delay-based PUFs.

A. Analog Electronic PUFs

These type of PUFs mainly include ICID [12], the coating PUF [15], Silicon Nano key [14], LC-PUF [9] and power grid PUFs [14].

1) ICID

Lofstrom *et al.* [12] proposed a unique IC identification (ICID) for each manufactured IC by utilizing the variations in the drain currents. This ICID method has less number of CRPs and has limited IDs (It is a weak PUF). So, it is not secure since adversaries can exhaustively read out the ID and “clone” an IC. ICID PUFs is also known as Vt (Threshold Voltage) PUFs. Figure.1. shows [12] basic block diagram of ICID block

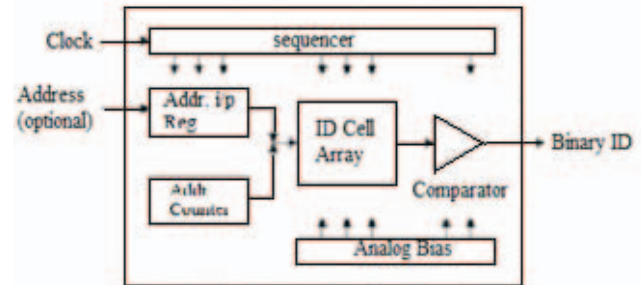


Figure.1. Block diagram of ICID block

2) Coating PUFs

Tuyls *et al.* [15] suggested a new PUF, coating PUF in which the protective coating is opaque and chemically neutral; these PUFs are resistant against invasive attacks. The applied voltage with a particular frequency and amplitude to the sensors is the challenge and the measured capacitance which is transformed into a bit string as an ID is response.

3) Silicon Nano key

Puntin *et al.* introduced the silicon Nano key [14]. The variance of the electrical parameters of small sized MOS transistors is used in the silicon Nano-keys, particularly the V_t (Threshold Voltage), to use in PUF with a CRP authentication scheme.

4) LC PUFs

The LC PUF was proposed in [9]. The Capacitor component is built as a tiny (about 1mm^2) glass plate along with a metal plate on all sides. The Inductor component, L is serially connected with a metal coil. Together these L and C components form a passive LC circuit. Even for equally manufactured circuits, the peak of the resonance frequency will be slightly different because of manufacturing process variations.

5) Power Grid PUFs

In a chip, depending on the resistance fluctuations in the power grid, a new PUF design was proposed by Helinski *et al.* [14]. It was observed that the random manufacturing process variations affect the voltage drops and equivalent resistances. Figure.2. shows circuit for generation of the signature using power grid.

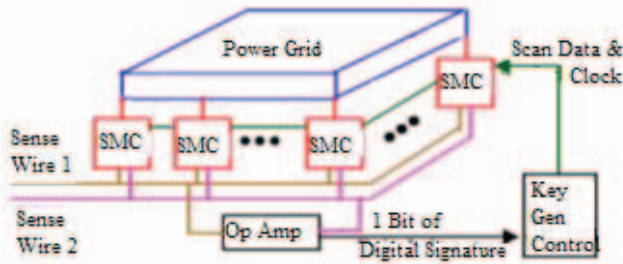


Figure.2. Circuit for generation of the signature using power grid.

B. Memory-Based PUFs

These types of PUFs mainly includes SRAM PUF [8], Butterfly PUF [11], MECCA PUF [10] etc.

1) Static RAM PUF (SRAM PUF)

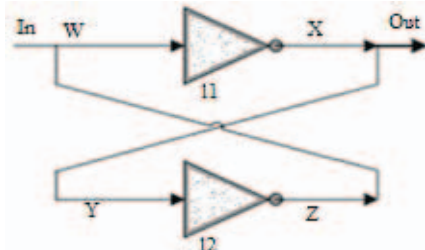


Figure.3. Basic structure of SRAM PUF

The SRAM PUF [8] comprises of a number of memory units. As shown in Figure.6, two cross-coupled inverters form

a memory unit. It has two stable states (0 and 1). The SRAM PUF is suitable only for the FPGAs which support the memories that do not need initialization. The error rate is relatively high for this type of PUF. If the internal cell layout is too symmetrical, the reliability also becomes low. It becomes prone to environmental noise, temperature changes, and power supply transients.

2) Butterfly PUF

Butterfly PUF is a modified version of SRAM PUF. Guajardo *et al.* proposed this PUF in [11]. In the butterfly PUF, an unstable cross-coupling circuit is used. Instead of inverters in SRAM PUF, a butterfly PUF uses a latch or flip-flop as shown in Figure.4. The advantages of this design are that it uses only D-Flip flops which are ubiquitous in FPGAs as well as in general design processes. One disadvantage is that it requires extra care to route, due to the constraints of FPGA routing.

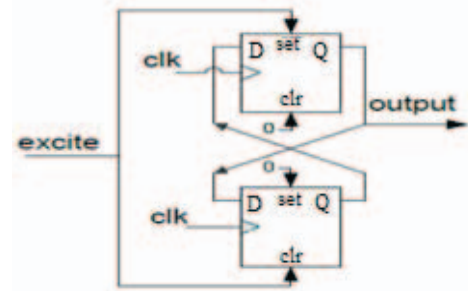


Figure.4. Butterfly PUF

3) MECCA PUFs

The SRAM PUF is open to the attack of exhaustive readout. To overcome this issue, MECCA PUF [10] was proposed by Krishna *et al.* In this PUF design, the write pulse duration is used as the challenge, and writes failures in memory cells would generate PUF response. But, MECCA PUF is only suitable for embedded memory of specific structures.

C. Delay-Based PUFs

These types of PUFs mainly include Arbiter PUF [5], Anderson PUF [1], Ring Oscillator PUF [3, 7], memristor PUF [6], Thyristor based PUF [4] etc.

1) Arbiter PUFs

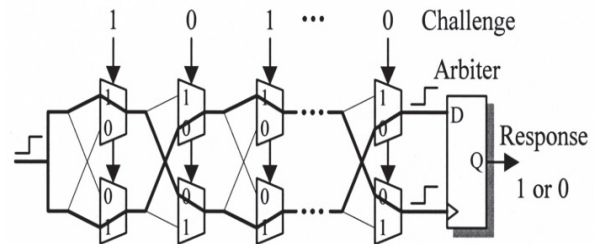


Figure.6. Basic Arbiter PUF

The arbiter PUF is another well-studied design, published in 2004 [5]. In the general sense, an arbiter PUF sets up a set of closely matched race tracks with an arbiter at the end to decide which signal reached the end first—typically this is a D Flip-Flop with one signal attached to the clock pin and another attached to the data pin. The basic arbiter PUF is as shown in Figure 6.

2) Ring Oscillator PUF

Ring oscillator (RO) is an earliest and mature classes of delay-based silicon PUFs, first introduced in [3, 7], and it is popular PUF designs on FPGAs, since absolute symmetry is not necessary to create an oscillator, and the error associated with making a single measurement is amortized across many oscillator cycles.

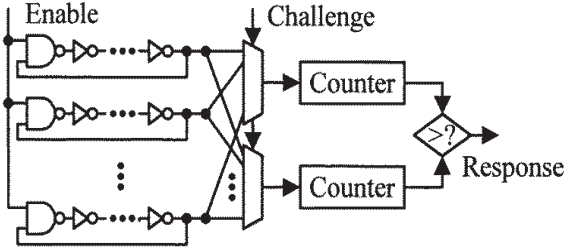


Figure.7. RO PUF with Differential configuration

The "differential" measurement with RO PUF has been shown to give better results than the basic RO design. A typical RO PUF with such a configuration is shown in Figure 7.

3) Anderson PUF

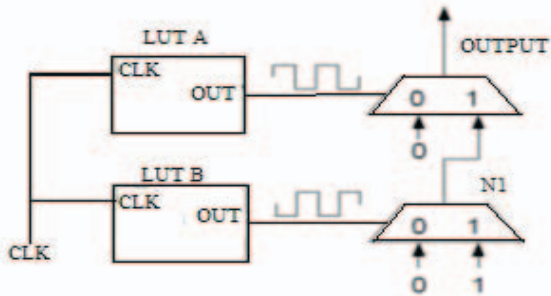


Figure.8. Structure of Anderson PUF

The Anderson PUF [1] is the first PUF designed for implementation on FPGAs. Unlike many PUFs designed for FPGAs, it uses the carry chain multiplexers present in certain FPGA components. A simplified depiction of Anderson's PUF is shown in Figure 8.

4) Memristor PUF

Memristors are electrical elements that are used to relate magnetic flux linkage and charge. A memristive PUF is given in [6] that use variability within the SET time of the memristors. A Memristive PUF cell (1-bit) is shown in Figure.9.

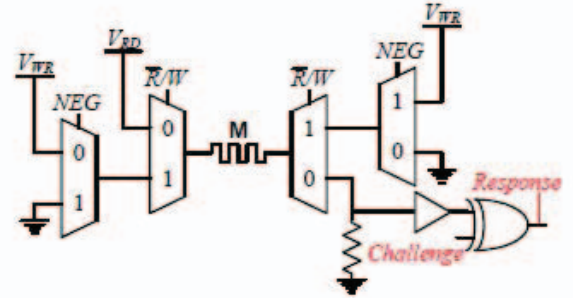


Figure.9. A memristive PUF cell (1 Bit)

5) Thyristor based PUF

A new PUF based on thyristor sensors was introduced by Chuang Bai et al. in [4] and the Figure.10 shows the architecture and basic operation of the thyristor-based PUF. It comprised of five circuit elements, and they are the thyristor-based sensors, Time Difference Amplifier (TDA), voting mechanism, Time Difference Comparator (TDC), and diffusion algorithm circuit. The challenge to the thyristor-based PUF is a set of select line bits, and the response is the output 0/1 bit. This is known as the CRPs.

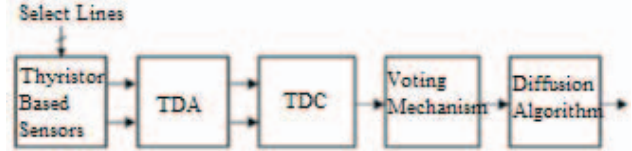


Figure.10. Basic operation of Thyristor based PUF

The thyristor sensors are made up of thyristor delay elements. These thyristor-based sensors catch the inherent manufacturing process variations. A set of sensors is made to be identical. Two slightly different delay time values are produced by each sensor depending on discharge current of delay elements. The time difference amplifier circuit amplifies the slight delay time difference from the selected sensors. The time difference comparator circuit compares the output delay time values from TDA to produce 0/1 bit ID for the IC. The voting mechanism is used to increase the stability of the ID bit produced by sampling the output of the TDC many times. The diffusion algorithm circuit ensures the uniform statistical distribution according to certain diffusion algorithms.

III. PROPOSED WORK

A new Silicon PUF can be designed as shown in the flow diagram Figure.11. The delay can be created in simulated stage by comparing the response of random polynomial equation and the multi-core device output. From this delay, a golden signature can be generated. So by using the inherent process variations and without excess extra hardware, the signature can be produced and authorization is possible by making use of this signature or ID.

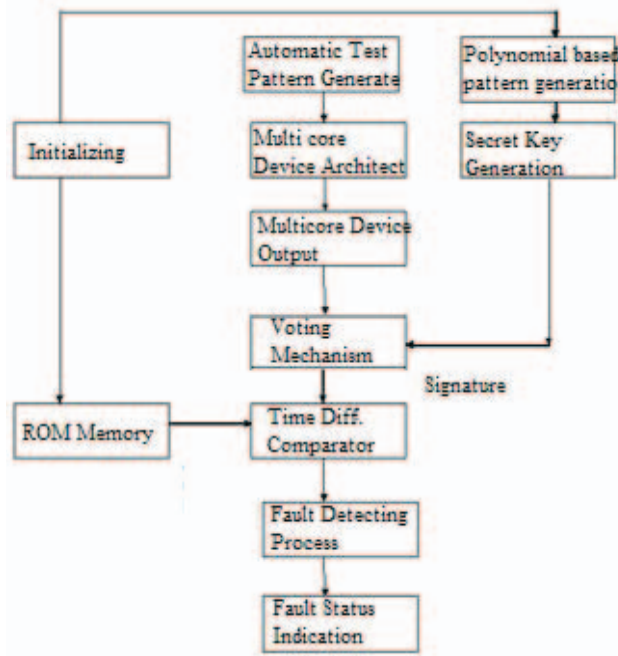


Figure.11. Flow Diagram of Proposed PUF System

IV. SUMMARY AND CONCLUSION

This paper provides a detailed study of different types of Silicon PUFs. This paper also proposes design of a new delay based Silicon PUF system. The proposed system can be explored further to utilize the low hardware requirements of the new design.

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